(Please write your Enrollment Number) Enrollment No. 0160117 2021

MID-TERM EXAMINATION
(Course Name: B.Tech. CSE-AI) (Semester: 5th)
(October, 2023) OFF LINE mode

Subject Code: BAI-305	Subject: Deep Learning
Time : 1 ½Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

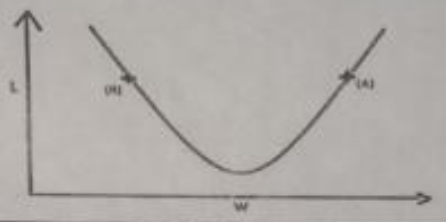
Q1 (2.5*4)

(a) Explain the purpose of thresholding logic in a MP-Neuron?

(b) Differentiate between MP-Neuron and Perceptron.

(c) Explain the relationship between partial derivative and chain-rule with an example.

(d) Consider the following curve of Loss (L) against weight parameter (w). There are two points on the curve which are highlighted as A and B. What will be the sign of the slope at these points? What changes do we need to make in the weight updation equation, if we apply gradient descent algorithm at points A and B?



Q2 (Attempt any Two Parts) UNIT-1 (5,5)

(a) Proof that perceptron learning algorithm will converge to an optimal solution?

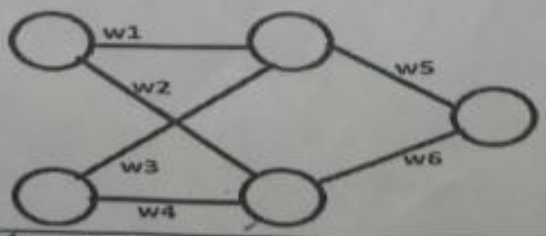
(b) Consider a three-input neuron with sigmoid function whose weights (w1, w2, w3) are initialized as (0.2, 0.5, 0.8) with objective function as squared error and bias as 0.3. What are the values of (w1, w2, w3) on applying gradient descent algorithm after two iterations with learning rate as 0.6 for following two data samples as (1,0,1) and (0,1,1) with output labels as 0 and 1, respectively.

(c) Is single-hidden neural network sufficient to solve all the Boolean functions with n Boolean inputs? If so, then what is need of using multi-layer neural network?

Q3 (Attempt any Two Parts) UNIT-2 (5,5)

(a) Explain the under-fitting and over-fitting with help of a diagram. Why over-fitting is a matter of concern in comparison to under-fitting?

(b) Consider the following artificial neural network of single hidden layer with initial values of (w1, w2, w3, w4, w5, w6) as (0.5, 0.2, 0.8, 1, 0.5, 0.5). The bias value to each neuron of hidden layer is 0.2 and to neuron of output layer is 0.3. The activation function is sigmoid and cost function is squared error function with gradient descent algorithm. Input sample values are (0.4, 0.8) with output label as 1. For the given weights and input, what is the predicted value by this network after applying the forward pass? What are the updated weights of these network after performing single backpropagation?



(c) Explain the following techniques in detail:
(i) L1 and L2 regularization
(ii) Dropout
(iii) Data Augmentation